Case Study 8: Password Policy Validation Engine

1. Problem Statement

Check a sample password against rules for minimum length, uppercase letter, lowercase letter, digit, and special character. Report every failed rule.

2. Objectives

Process strings using loops.

Check character categories.

Report multiple validation failures.

3. Assumptions

Minimum length is 8 characters.

A valid password must contain at least one uppercase letter, lowercase letter, digit, and special character.

Whitespace is not accepted as a special character.

4. Algorithm

1. Start.

2. Input a sample password.

3. Check its length.

4. Loop through each character to identify uppercase, lowercase, digit, and special character.

5. Collect all failed rules.

6. Display whether the password passes and list failures.

7. Stop.

5. Flowchart (text form)

6. Pseudocode

```
START
INPUT password
Create flags for uppercase, lowercase, digit, special character
FOR each character: update the matching flag
Create empty list failures
IF length < 8: add length failure
IF any category flag is false: add corresponding failure
IF failures is empty: display valid; ELSE display invalid and failures
STOP
```

7. Complete Python Source Code

```
password = input("Enter a sample password to check: ")
```



```
has_upper = False
has_lower = False
has_digit = False
has_special = False
```

```
    for ch in password:
     if ch.isupper():
has_upper = True
elif ch.islower():
has_lower = True
elif ch.isdigit():
has_digit = True
elif not ch.isspace():
has_special = True

failures = []
    if len(password) < 8:
failures.append("At least 8 characters are required.")
    if not has_upper:
failures.append("Add at least one uppercase letter.")
    if not has_lower:
failures.append("Add at least one lowercase letter.")
    if not has_digit:
failures.append("Add at least one digit.")
    if not has_special:
failures.append("Add at least one special character.")
    if len(failures) == 0:
     print("Password passes the sample policy.")
    else:
     print("Password does not pass the sample policy.")
     for reason in failures:
     print("-", reason)
```

8. Test Case and Expected Output

Input: hello123

Password does not pass the sample policy. - Add at least one uppercase letter. - Add at least one special character.

9. Conclusion

The program checks a string against multiple password rules and explains each failed condition. It is a classroom example, not a complete real-world security checker.